

Energy-efficient edge inference for implantable devices: Mechanisms and Interventions

Specific Aims

Aim 1: Advance energy-efficient edge inference for implantable devices via a fabrication-focused strategy [1].

Aim 2: Advance energy-efficient edge inference for implantable devices via a fabrication-focused strategy [2].

Aim 3: Advance energy-efficient edge inference for implantable devices via a fabrication-focused strategy [3].

Background

This proposal addresses energy-efficient edge inference for implantable devices, a problem with substantial significance [4]. Prior work has established a partial understanding, yet key gaps remain. We will assemble a multidisciplinary team with complementary expertise. Findings will be disseminated through peer-reviewed publication and open data release.

Approach

This proposal addresses energy-efficient edge inference for implantable devices, a problem with substantial significance. Prior work has established a partial understanding, yet key gaps remain. Our preliminary data suggest a tractable path toward measurable improvement. Rigorous, pre-registered analyses will guard against bias and support reproducibility.

Innovation

Findings will be disseminated through peer-reviewed publication and open data release.

Investigator Context

The PI's first independent award; strong mentorship is in place. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Budget: personnel \$216,982; equipment \$57,938; travel \$15,751; indirect costs \$45,726. Total requested support is \$336,397 over 2 years at a R2 institution.

As one source notes, "Prior work has established a partial understanding, yet key gaps remain." [1]

References

1. [1] <https://doi.org/10.1056/NEJMoa2034577>
2. [2] <https://doi.org/10.1109/TPAMI.2016.2577031>
3. [3] <https://doi.org/10.1109/TPAMI.2016.2577031>

4. [4] <https://doi.org/10.1001/jama.2016.9797>